

Proma Roy

Brooklyn, NY



heypromaroy



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Summary

Graduate cybersecurity student with 3+ years of professional experience in data analysis, documentation workflows, and database management. Strong interest in research-oriented secure systems, cloud computing, and applying software engineering and security principles to socially impactful, interdisciplinary projects.

Education

The City College of New York (CUNY)

Master of Science in Cybersecurity

New York, NY

Aug 2025 – Jun 2027 (Expected)

United International University

Bachelor of Science in Computer Science & Engineering

Dhaka, Bangladesh

Jan 2016 – Dec 2019

Academic & Technical Projects

Security Audit of Rust-Based E-Voting System

Nov 2025 – Dec 2025

Secure Systems Engineering, The City College of New York

- Conducted a comprehensive security audit of a Rust-based electronic voting system as part of Audit & Analysis.
- Evaluated adversarially modified codebases as part of the ThermoRust analyst team.
- Identified high-impact vulnerabilities across authentication, access control, cryptography, data privacy, availability, and build integrity.
- Applied STRIDE threat modeling, OWASP Top 10 and CWE mapping, static analysis, runtime fault injection, and binary/build pipeline inspection.
- Proposed concrete mitigation strategies aligned with secure-by-design and responsible disclosure principles.

Smart Thermostat — Secure CLI HVAC Prototype (Rust)

Sep 2025 – Oct 2025

Secure Systems Engineering, The City College of New York

- Built a secure, role-based smart thermostat CLI prototype in Rust with HVAC simulation, energy tracking, and hardened SQLite storage.
- Implemented Argon2id password hashing, session management with hashed session tokens, and account/session lockouts.
- Designed role-based access control flows (homeowner, guest, technician, admin).
- Integrated audit logging, tamper-evident hash-based integrity verification, and secure database PRAGMA settings.
- Developed integration tests covering authentication, database, and HVAC modules.

Professional Experience

Associate Vendor Coordinator

Mar 2022 – Feb 2025

Radiant Data Systems, Ltd. (AMN Healthcare Partner)

- Verified licenses and credentials of healthcare professionals, achieving consistent 100% audit accuracy.
- Maintained and validated databases to ensure accuracy, compliance, and traceability.
- Recognized for attention to detail, process integrity, and reliability.

Data Artist

Nov 2021 – Mar 2022

Adiva Catalyst Solution & Graphics

- Reviewed survey logic to ensure alignment with research and client objectives.
- Performed data verification and produced analytical visualizations using SPSS and Excel.
- Assisted with technical troubleshooting and quality assurance.

Junior Data Analyst

Jun 2020 – Jun 2021

a2i (ICT Division), Bangladesh

- Supported the COVID-19 Telehealth & Telemedicine Center project with data collection and validation.
- Maintained project documentation and coordinated with IT and healthcare stakeholders.
- Assisted with CRM data tracking and reporting accuracy.

Skills

System Security: Secure Systems Engineering, Threat Modeling (STRIDE), OWASP Top 10, CWE, Audit & Analysis

Programming & Data: Rust, Python, SQL, SQLite, Static Analysis, Integration Testing

Tools & Platforms: Git/GitHub, VSCode, SPSS, Excel, Documentation & Reporting

Certifications

Palo Alto Networks: Cybersecurity Fundamentals; Network Security Fundamentals; Endpoint Security; Cloud Security Fundamentals; Understanding Security Operations (Foundational)